

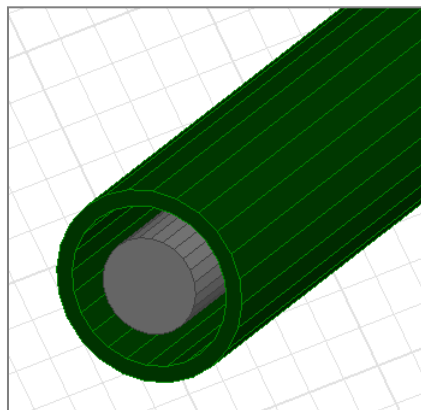
Workshop 2.4: 3D Electrostatic analysis

Release 2020R2

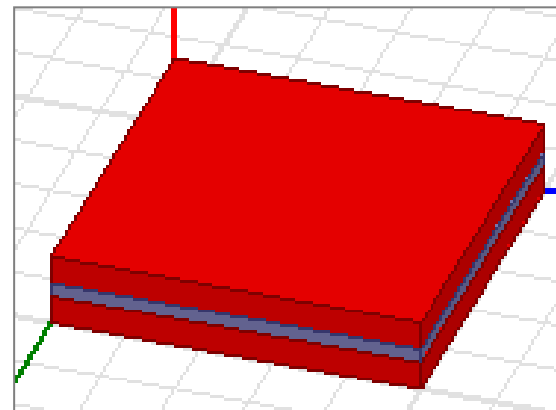


Overview

- Introduction to the Electrostatic Solver
 - This workshop introduces the Electro Static solver based on some simple examples. This solver is meant to solve the static electric field without current flowing in conductors (conductors are in electrostatic equilibrium). The conductors are considered perfect such that there is no electric field inside them
- The Workshop contains following two examples
 - Example1: **Cylindrical Capacitor**
 - In this example, we will determine the electric field distribution of coaxial cable based on the potential (or the charges) that are applied on each conductor. We will also evaluate the capacitance.
 - Example2: **Capacitance of a Planar Capacitor**
 - In this example we simulate a simple planar capacitor made of two parallel conducting plates



Example 1

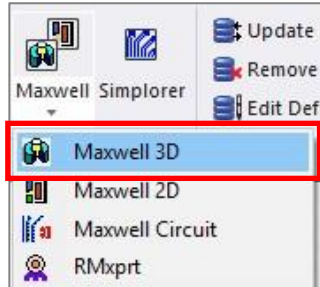


Example 2

Example 1: Cylindrical Capacitor - Model Setup

- Insert Design

- Select the menu item **Project** → **Insert Maxwell 3D Design**, or click on the icon in drop down list Maxwell on panel Desktop



- Set Solution Type

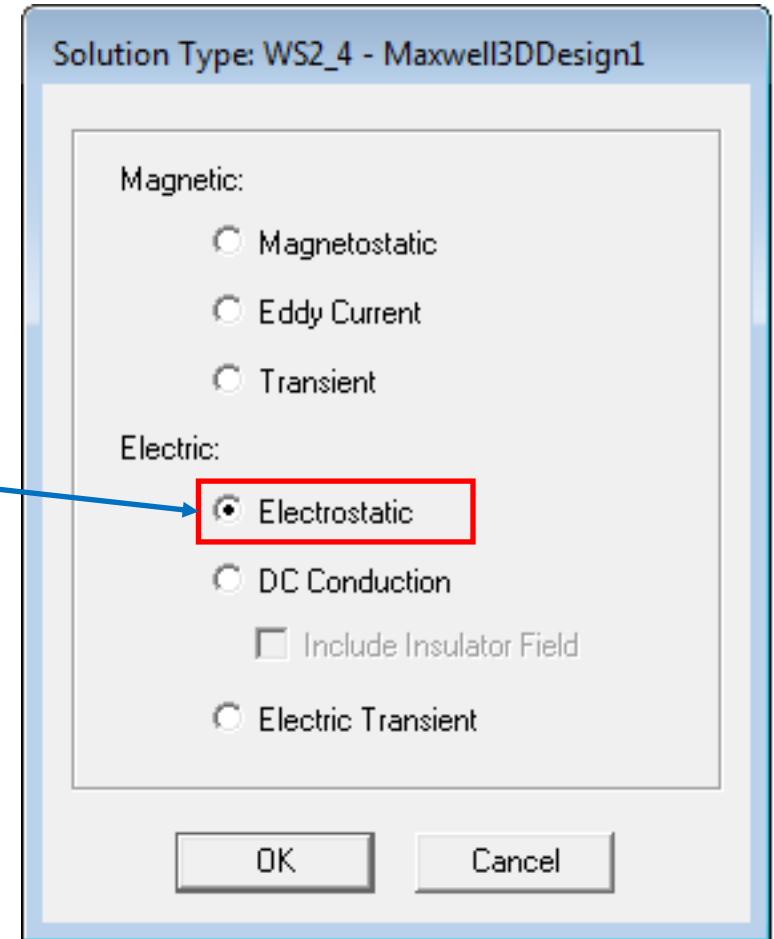
- Select the menu item **Maxwell 3D** → **Solution Type**
- Choose **Electric** → **Electrostatic**
- Click the OK button

- Set Model Units

- Select the menu item **Modeler** → **Units**
 - Select units: mm (millimeters) and press OK

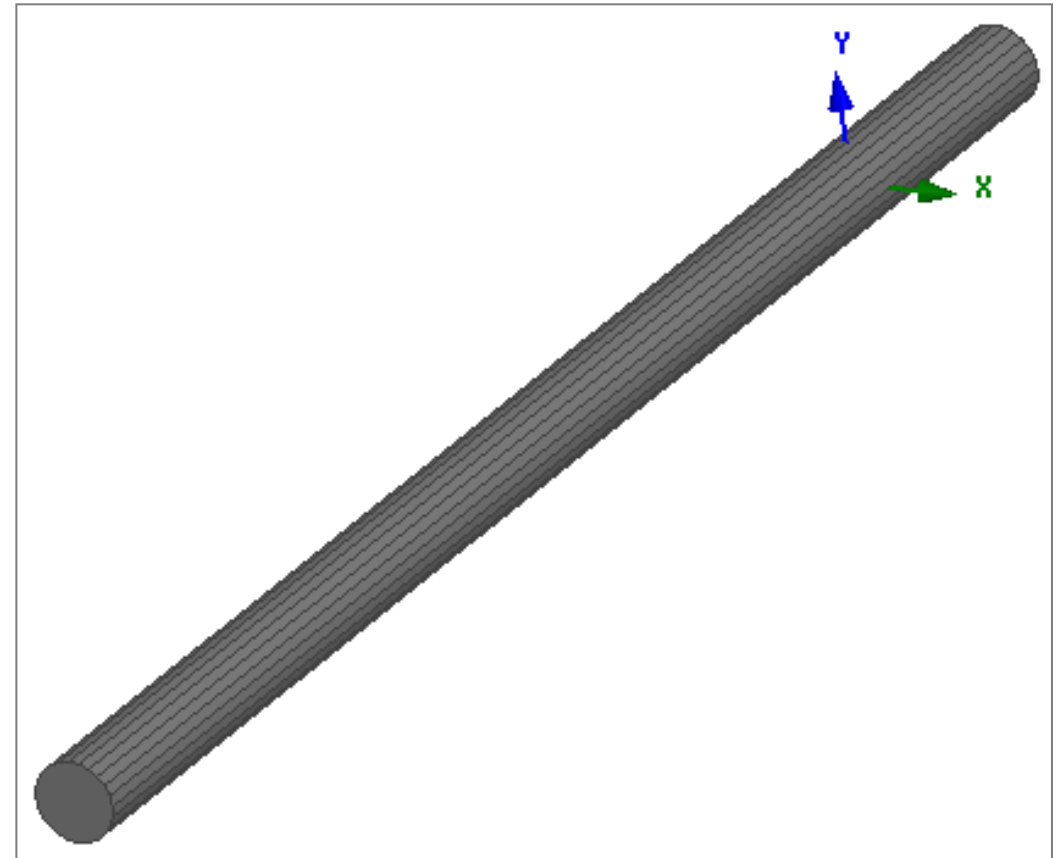
- Set the default material

- Using the 3D Modeler Materials toolbar, choose Select
- In Select Definition window,
 - Type **Copper** in the Search by Name field and press OK



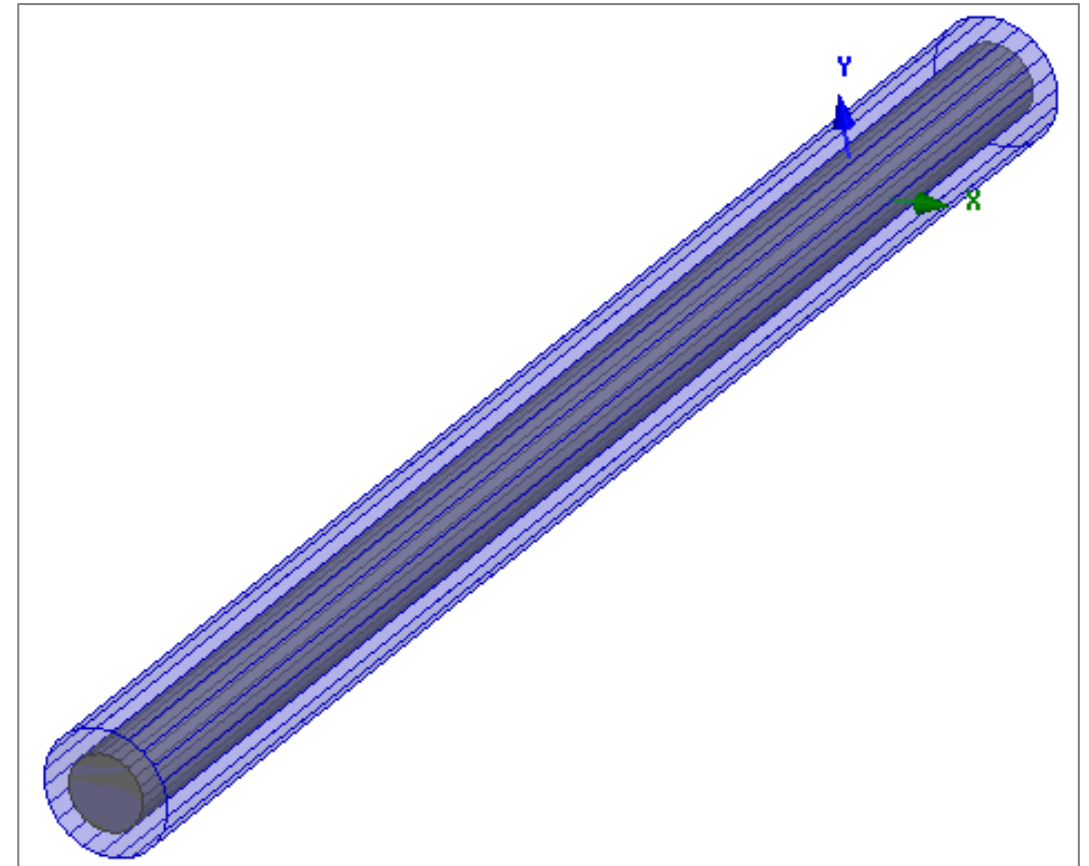
Create Model

- Create Regular Polyhedron
 - Select the menu item **Draw** → **Regular Polyhedron**
 - Using the coordinate entry fields, enter the center of the base
X: 0, Y: 0, Z: -4, Press the Enter key
 - Using the coordinate entry fields, enter the radius
dX: 0.6, dY: 0, dZ: 0, Press the Enter key
 - Using the coordinate entry fields, enter the height
dX: 0, dY: 0, dZ: 25, Press the Enter key
 - Number of Segments: **24**
 - Change the name of the Object to **Inner**
 - Set the color of the object to **Gray**



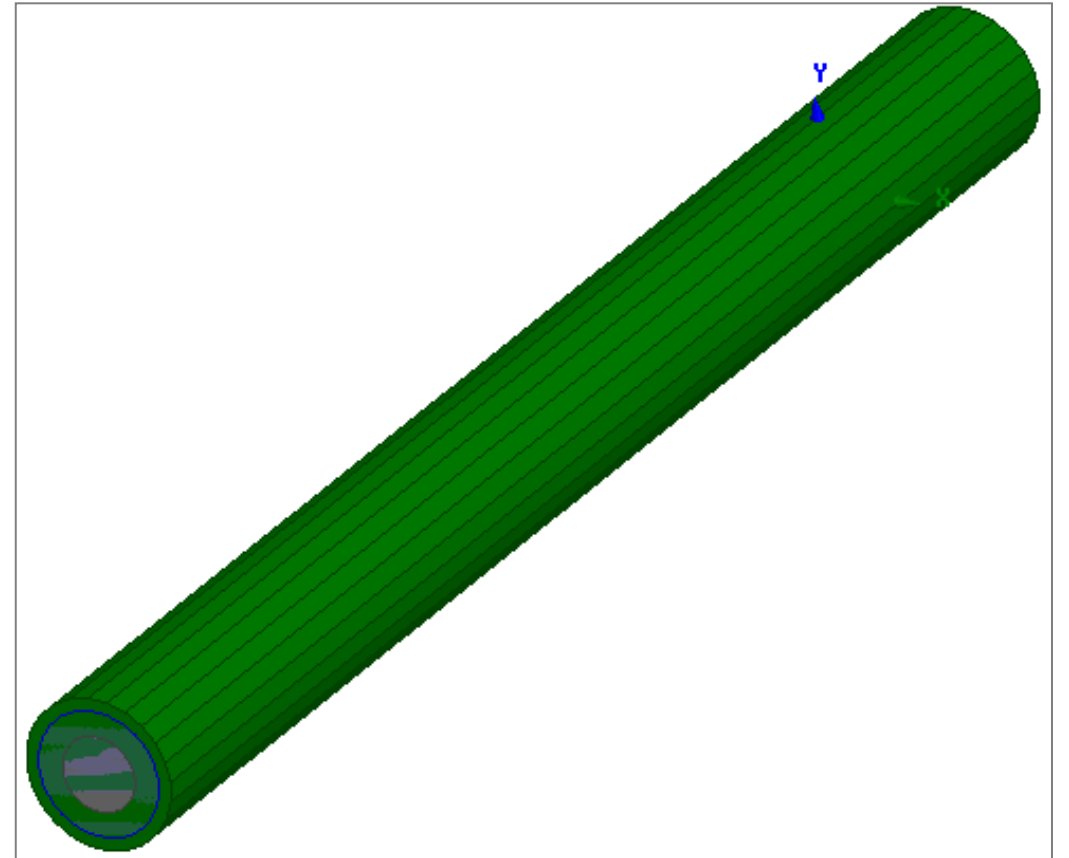
Create Model

- Create Second Polyhedron
 - Select the menu item **Draw** → **Regular Polyhedron**
 - Using the coordinate entry fields, enter the center of the base
X: 0, Y: 0, Z: -4, Press the Enter key
 - Using the coordinate entry fields, enter the radius
dX: 1, dY: 0, dZ: 0, Press the Enter key
 - Using the coordinate entry fields, enter the height
dX: 0, dY: 0, dZ: 25, Press the Enter key
 - Number of Segments: **24**
 - Change the name of the Object to **Gap**



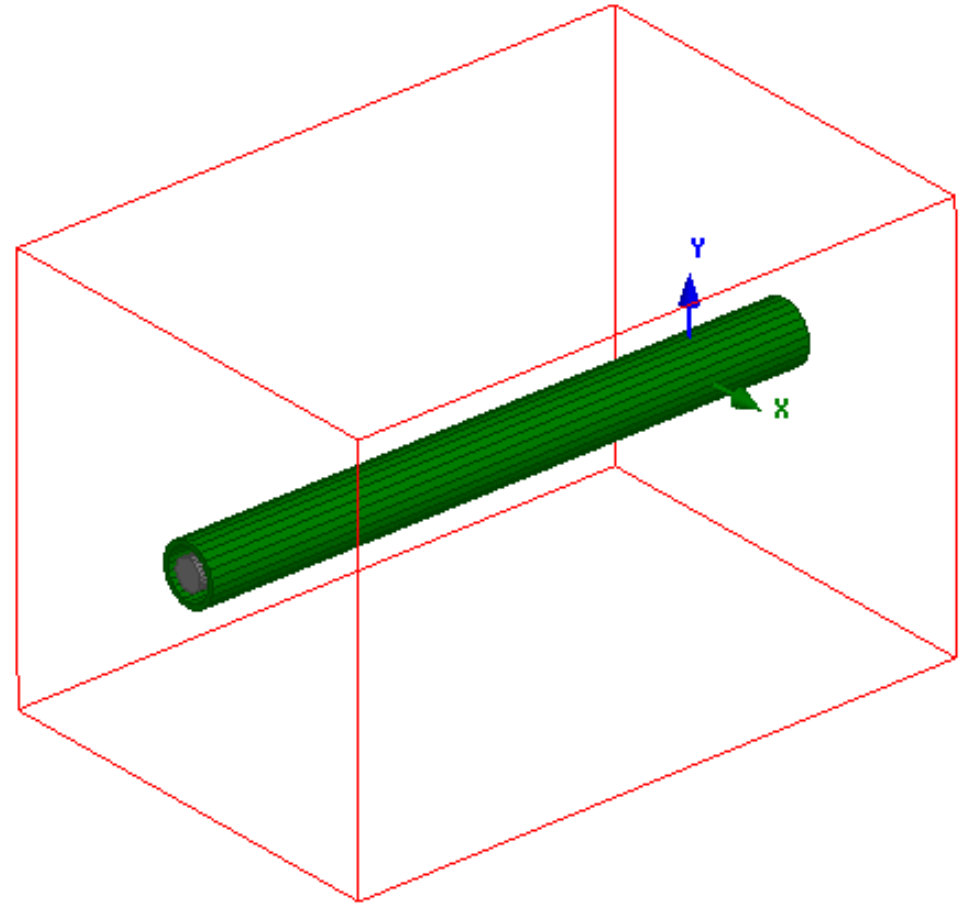
Create Model

- Create Third Polyhedron
 - Select the menu item **Draw** → **Regular Polyhedron**
 - Using the coordinate entry fields, enter the center of the base
X: 0, Y: 0, Z: -4, Press the Enter key
 - Using the coordinate entry fields, enter the radius
dX: 1.2, dY: 0, dZ: 0, Press the Enter key
 - Using the coordinate entry fields, enter the height
dX: 0, dY: 0, dZ: 25, Press the Enter key
 - Number of Segments: **24**
 - Change the name of the Object to **Outer**
 - Set the color of the object to **Green**



Create Model

- Subtract Objects
 - Press Ctrl and select the objects **Outer** and **Gap** from the history tree
 - Select the menu item, *Modeler* → *Boolean* → *Subtract*
 - Blank Parts: **Outer**
 - Tool Parts: **Gap**
 - Click the OK button
- Create Simulation Region
 - Select the menu item *Draw* → *Region* or click on the icon 
 - In Region window,
 - Pad individual directions: ☒ **Checked**
 - Padding Type: Percentage Offset
 - +X and -X = **300**
 - +Y and -Y = **300**
 - +Z and -Z = **0**
 - Press OK



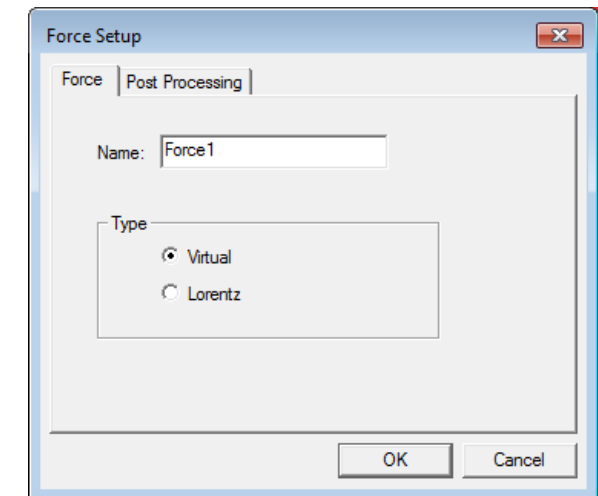
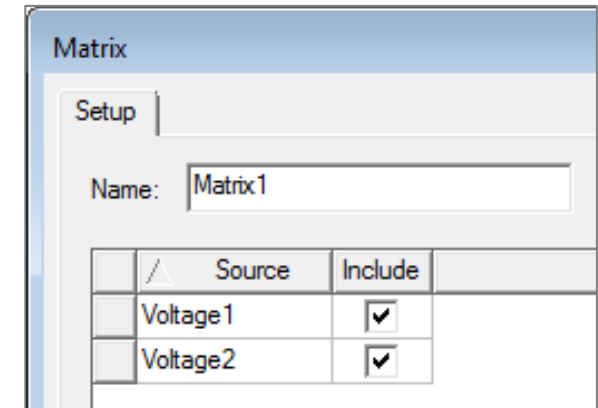
/ Excitations

- Assign Excitation for Inner
 - Select the object **Inner** from the history tree
 - Select the menu item **Maxwell 3D → Excitations → Assign → Voltage**
 - In Voltage Excitation window,
 - Set Value to **-1000 V**
 - Press OK
- Assign Excitation for Outer
 - Select the object **Outer** from the history tree
 - Select the menu item **Maxwell 3D → Excitations → Assign → Voltage**
 - In Voltage Excitation window,
 - Set Value to **1000 V**
 - Press OK

Note: Assuming an electrostatic equilibrium, the conductors are assigned as an equipotential voltage

/ Assign Parameters

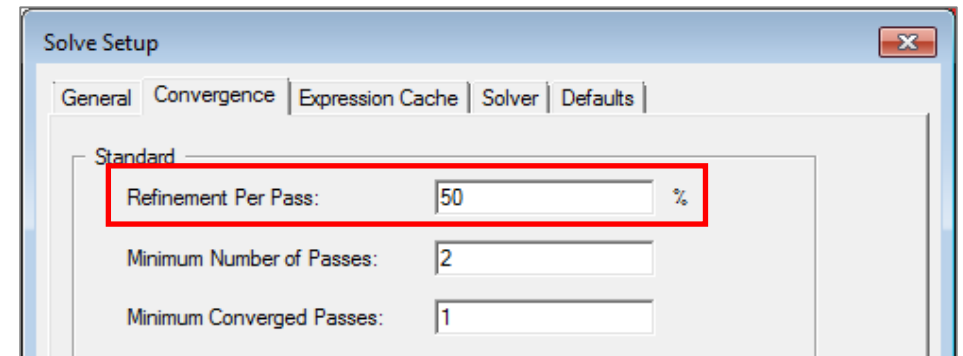
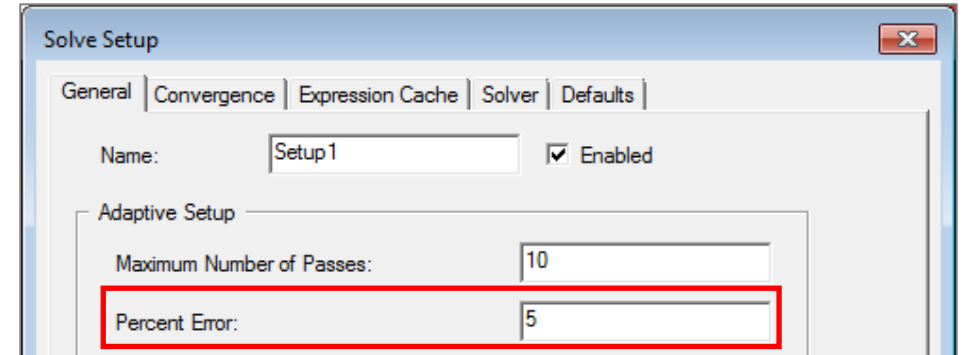
- Assign Matrix Parameter to Calculate Capacitance
 - Select the menu item **Maxwell 3D** → **Parameters** → **Assign** → **Matrix**
 - In Matrix window
 - Voltage1 and Voltage2
 - Include: ☒ Checked
 - Press OK
- Assign Force Parameter for Inner
 - Select the Object **Inner** from the tree
 - Select the menu item **Maxwell 3D** → **Parameters** → **Assign** → **Force**
 - In Force Setup window,
 - Type : **Virtual**
 - Press OK



Note: Parameters are assigned for calculation of capacitance value as well as the force applied to the inner armature.

Analyze

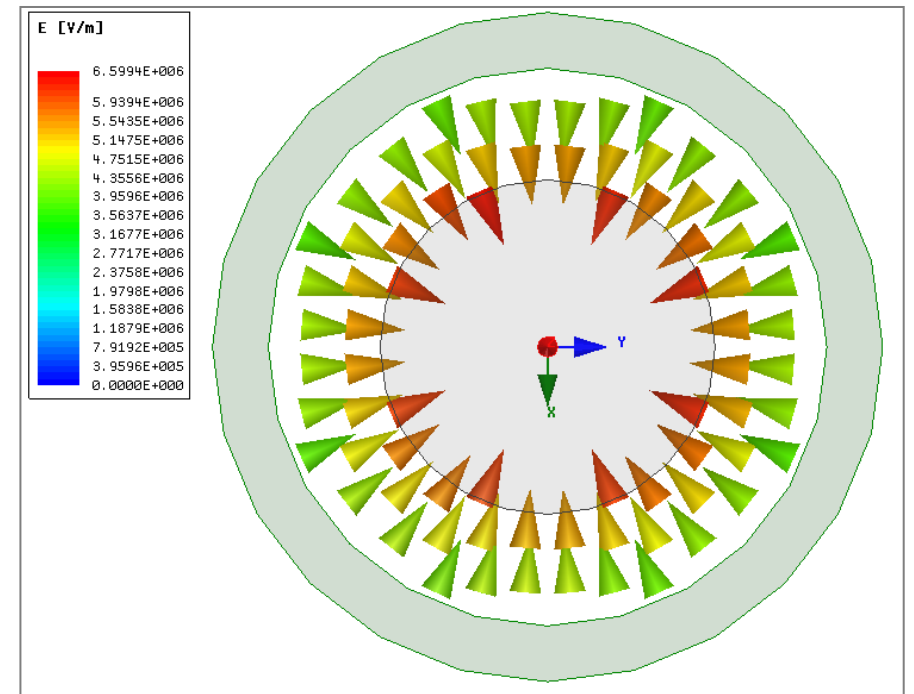
- To create an analysis setup:
 - Select the menu item **Maxwell 3D** → **Analysis Setup** → **Add Solution Setup**
 - Solution Setup Window:
 - **General Tab**
 - Percent Error: 5
 - **Convergence Tab**
 - Refinement Per Pass: 50%
 - Click the OK button
- To start the solution process:
 - In the Project Manager window **RMB on Setup1** → **Analyze**



Note: You will need to save your project before analyzing. Note that the project is always saved before the simulation starts.

Create Vector Plot

- To Create Vector Plot
 - Select the plane **Global:XY** from the history tree
 - Select the menu item **Maxwell 3D → Fields → Fields → E → E_Vector**
 - In Create Field Plot window
 - Press Done



Note: Default plot may not look as shown in image. Users can modify attributes of plot to make it look better. Double click on the legend to modify plot attributes.

View Results

- View Capacitance

- Select the menu item *Maxwell 3D* → *Results* → *Solution Data*
- Select the **Matrix** tab
- The analytical value of the capacitance per meter for an infinite long coaxial wire is given by the following formula:

$$C = 2\pi\epsilon_0 / \ln(b/a) \text{ (a and b being the inside and outside diameters)}$$

- The analytical value would be therefore 1.089e-10 F/m (a = 0.6mm, b = 1mm)
- In our project, then length of the conductor is 25 mm, therefore the total capacitance is 2.72 pF. We obtain a good agreement with the obtained result.

	Voltage1	Voltage2
Voltage1	2.7319	-2.7319
Voltage2	-2.7319	2.7319

View Results

- View Force
 - Select the **Force** tab

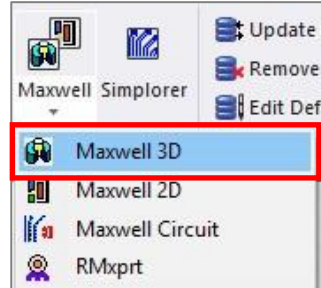
Profile	Convergence	Force	Torque	Matrix	Mesh Statistics
Parameter:		Force1	Force Unit:		newton
Pass:		2			
	F(x)	F(y)	F(z)	Mag(F)	
Total	-7.6427E-006	6.3109E-007	2.5412E-007	7.6729E-006	

Note: the force is practically zero due to the symmetry in the geometry

Example 2: Planar Capacitor - Model Setup

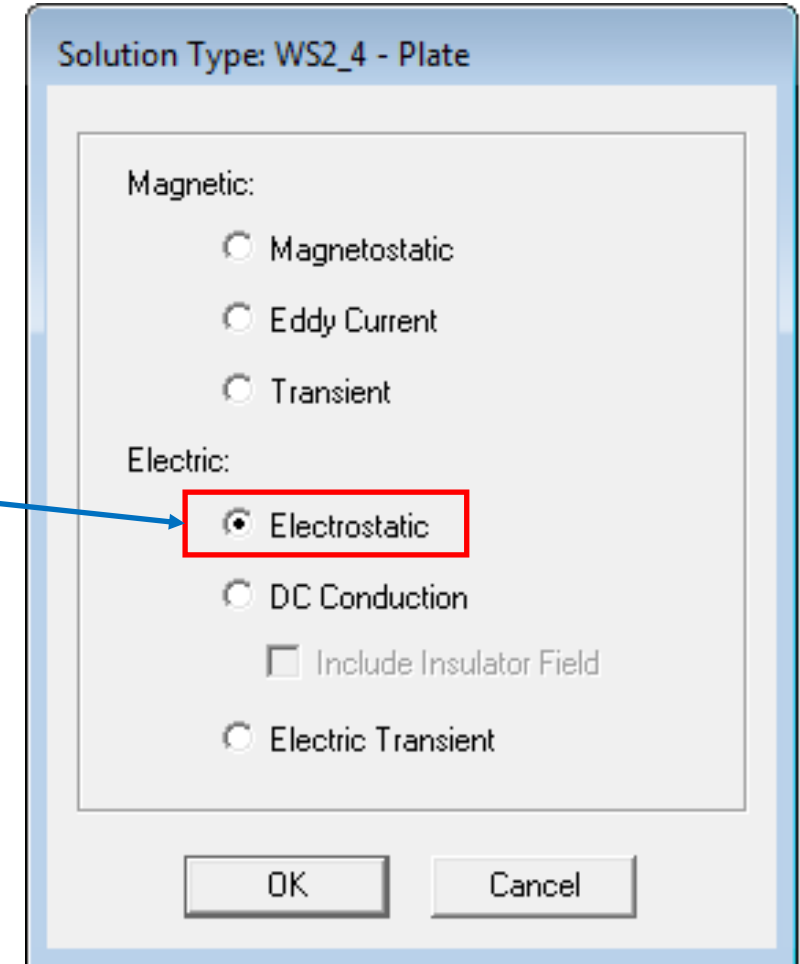
- Insert Design

- Select the menu item Project → Insert Maxwell 3D Design, or click on the icon in drop down list Maxwell on panel Desktop
- Rename the Design as Plate



- Set Solution Type

- Select the menu item *Maxwell 3D* → *Solution Type*
- Choose *Electric* → *Electrostatic*
- Click the OK button



- Set Model Units

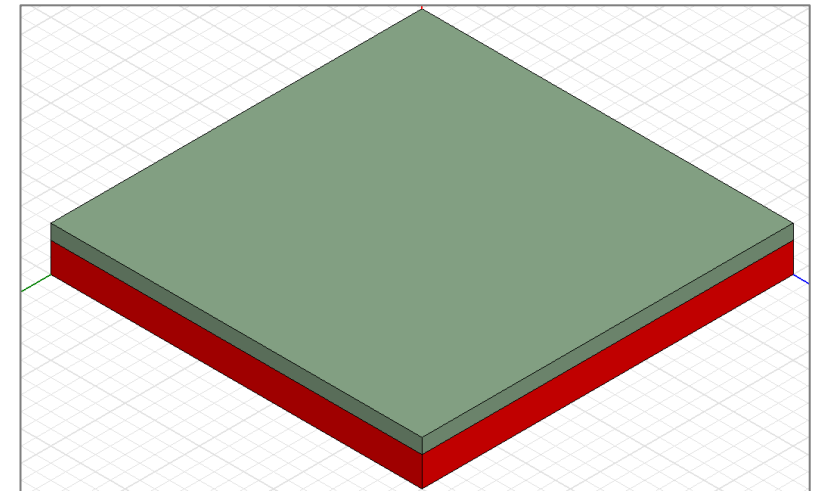
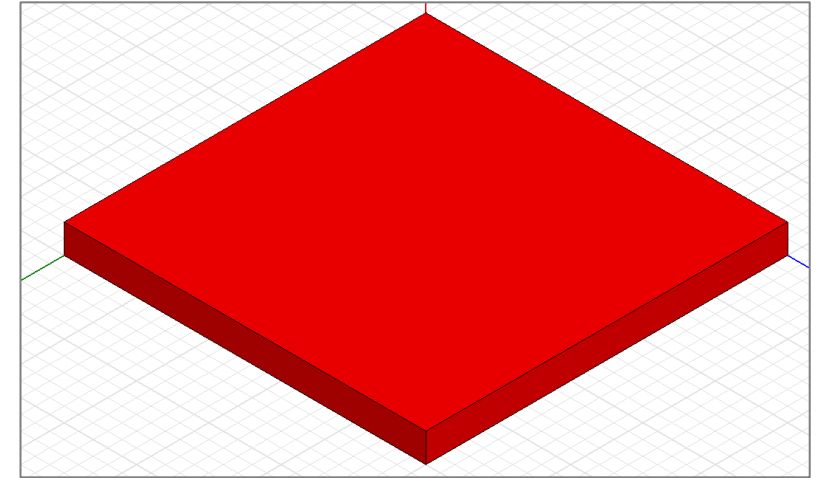
- Select the menu item *Modeler* → *Units*
 - Select units: **mm** (millimeters) and press OK

- Set the default material:

- Using the 3D Modeler Materials toolbar, choose Select
 - Type **Pec** (**Perfect Conductor**) in the Search by Name field
 - Press OK

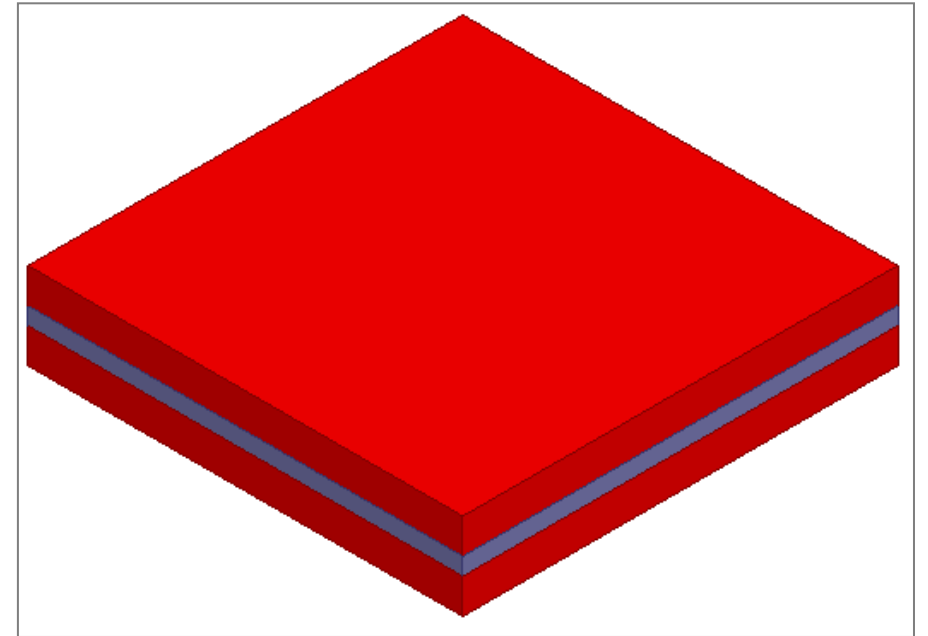
Create Model

- Create Box
 - Select the menu item *Draw* → *Box*
 - Using the coordinate entry fields, enter the box position
X: 0, Y: 0, Z: 0, Press the Enter key
 - Using the coordinate entry fields, enter the opposite corner
dX: 25, dY: 25, dZ: 2, Press the Enter key
 - Change the name of the Object to *DownPlate* and color to *Red*
- Create Gap
 - Select the menu item *Draw* → *Box*
 - Using the coordinate entry fields, enter the box position
X: 0, Y: 0, Z: 2, Press the Enter key
 - Using the coordinate entry fields, enter the opposite corner
dX: 25, dY: 25, dZ: 1, Press the Enter key
 - Change the name of the Object to *Gap* and material to *Vacuum*



Create Model

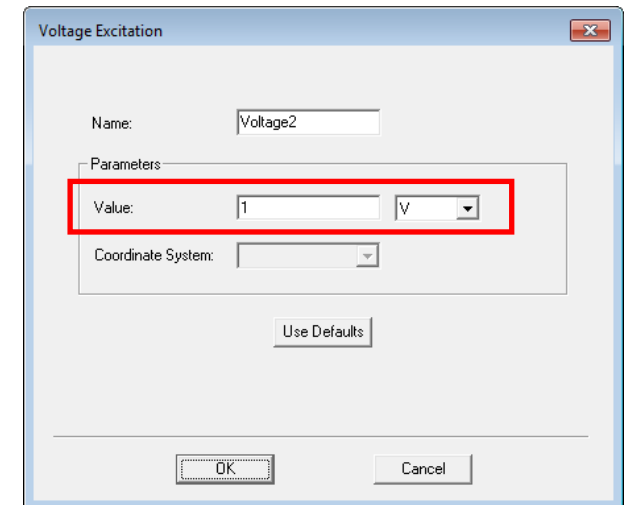
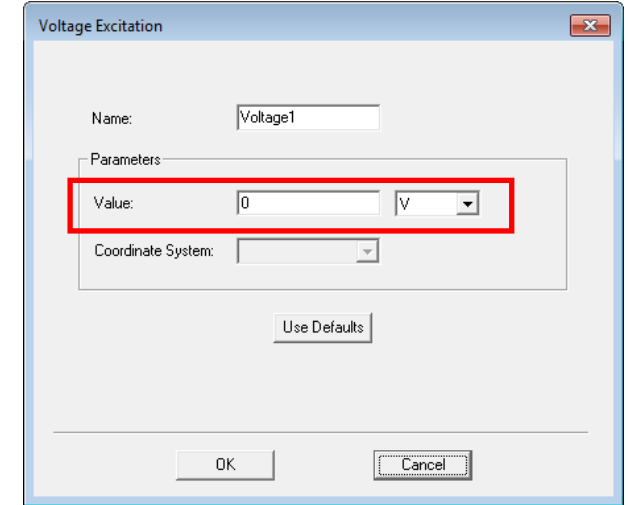
- Create Second Box
 - Select the menu item **Draw** → **Box**
 - Using the coordinate entry fields, enter the box position
X: 0, Y: 0, Z: 3, Press the Enter key
 - Using the coordinate entry fields, enter the opposite corner
dX: 25, dY: 25, dZ: 2, Press the Enter key
 - Change the name of the Object to **UpPlate**
 - Change the colors of the objects if needed



Note: It is not needed to draw the plates; it is possible in this case just to apply a voltage to the surface of the gap.

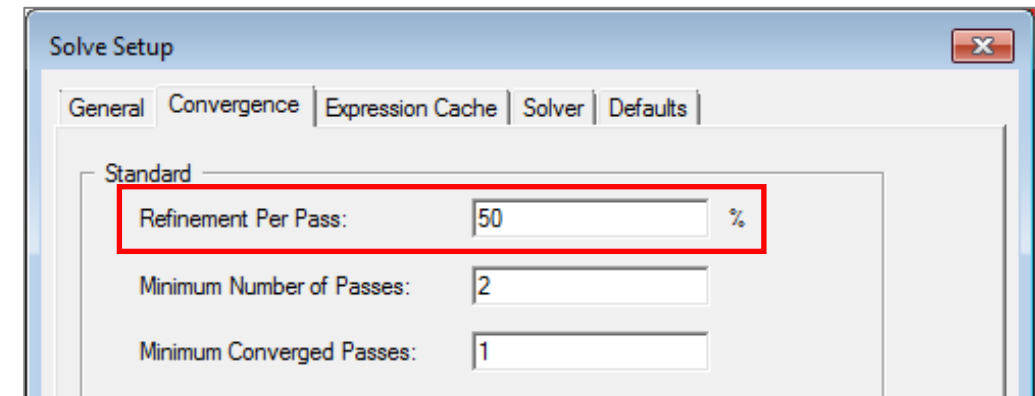
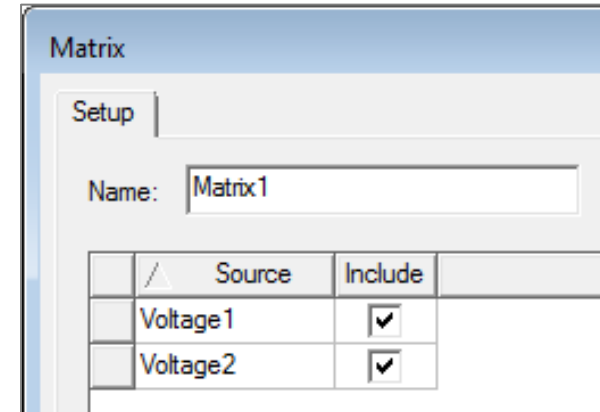
Excitations

- Assign Excitation for DownPlate
 - Select the object **DownPlate** from the history tree
 - Select the menu item **Maxwell 3D → Excitations → Assign → Voltage**
 - In Voltage Excitation window,
 - Set Value to 0 V
 - Press OK
- Assign Excitation for UpPlate
 - Select the object **UpPlate** from the history tree
 - Select the menu item **Maxwell 3D → Excitations → Assign → Voltage**
 - In Voltage Excitation window,
 - Set Value to 1 V
 - Press OK



Analyze

- Assign Matrix Parameter to Calculate Capacitance
 - Select the menu item **Maxwell 3D → Parameters → Assign → Matrix**
 - In Matrix window
 - Voltage1 and Voltage2
 - Include: ☒ Checked
 - Create an analysis setup:
 - Select the menu item **Maxwell 3D → Analysis Setup → Add Solution Setup**
 - Solution Setup Window:
 - **Convergence Tab**
 - Refinement Per Pass: 50%
 - Click the OK button
- Start the solution process:
 - In the Project Manager window **RMB on Setup1 → Analyze**



View Results

- View Capacitance

- Select the menu item *Maxwell 3D* → *Results* → *Solution Data*
- Select the Matrix tab
- The analytical value of the capacitance per meter for an infinite long coaxial wire is given by the following formula:

$C = A / d * \epsilon_0$ (A is the area of the plate and d is the thickness of the dielectrics)

- Using this formula, we obtain 5.53 pF. This value matches the value obtain using the finite elements

The screenshot shows the 'Matrix' tab in the 'Results - Solution Data' dialog box. The 'Parameter' is set to 'Matrix1' and the 'Type' is 'Capacitance'. The 'Pass' is set to '2' and the 'Capacitance Units' are 'pF'. Below the settings is a table with two columns: 'Voltage1' and 'Voltage2'. The table contains two rows: 'Voltage1' with values 5.5339 and -5.5339, and 'Voltage2' with values -5.5339 and 5.5339. The value 5.5339 in the first row is highlighted with a red box.

	Voltage1	Voltage2
Voltage1	5.5339	-5.5339
Voltage2	-5.5339	5.5339

Saving the Project

- This completes the workshop
- Save the file with the name **Workshop_2_4** in the working folder



End of Presentation